Final Exam Problems

MATH 598 / 784 · High-Dimensional Probability

November 23–27, 2026

Prepare all eight problems. The oral exam consists of two randomly selected problems presented at the blackboard in **Burnside 925**. There are **20 minutes for the entire examination**, within your booked 30-minute appointment. Arrive at the start of your appointment. The instructor may ask you to explain steps, justify assumptions, or discuss the cases specified in the problems. You may cite the facts explicitly supplied below without proving them.

Each of the two presentations is worth up to 25 points. A replacement problem costs 5 points; the total score is at least zero. For the complete procedure, rubric, and booking information, see the final exam page.

Throughout, $C, c > 0$ denote absolute constants, whose values may change. All logarithms are natural. Write $B_p^n = \{x \in \mathbb{R}^n : \|x\|_p \le 1\}$, and, for a nonempty bounded set T in a Euclidean space, write $w(T) = \mathbb{E}\sup_{t\in T}\langle g, t\rangle$ for standard Gaussian g. Covering numbers have centers in the set being covered.

1. Gaussian quadratic forms and effective dimension

Let A be a nonzero real symmetric $n \times n$ matrix, let $g \sim N(0, I_n)$, and set $Q = g^{\mathsf{T}}Ag - \operatorname{tr} A$.

1. Diagonalize A and compute $\mathbb{E}e^{\theta Q}$. Show that
$$\log \mathbb{E}e^{\theta Q} \le C\theta^2\|A\|_F^2 \quad \text{if } |\theta| \le c/\|A\|_{\mathrm{op}}.$$
2. Optimize the exponential Markov bound to prove, for $t > 0$,
$$\mathbb{P}\{|Q| \ge t\} \le 2\exp\left[-c\min\left\{\frac{t^2}{\|A\|_F^2}, \frac{t}{\|A\|_{\mathrm{op}}}\right\}\right].$$
3. Suppose $A \succeq 0$ and define $r_2(A) = (\operatorname{tr} A)^2/\|A\|_F^2$ and $r_\infty(A) = \operatorname{tr} A/\|A\|_{\mathrm{op}}$. Express the relative-error bound at $t = \varepsilon \operatorname{tr} A$ in terms of these quantities. Explain the two regimes using an orthogonal projection of rank r, and check that $r_\infty(A) \le r_2(A) \le \operatorname{rank} A$.

Hint. For $|u| \le 1/2$, $-\log(1-u) - u \le Cu^2$. The argument must allow negative eigenvalues in parts 1 and 2.

2. One random map for a union of subspaces

Let $V_1, \dots, V_M$ be linear subspaces of $\mathbb{R}^n$, each of dimension at most k, where $1 \le k \le n$ and $M \ge 1$. Let G be an $m \times n$ matrix of independent $N(0,1)$ entries. Fix $\varepsilon, \rho \in (0,1)$ and put $T = \bigcup_{j=1}^M V_j$.

1. For a fixed d-dimensional subspace W, $d \ge 1$, use a $1/4$-net of its unit sphere and Problem 1 to prove
$$\mathbb{P}\left\{\sup_{x\in W\cap S^{n-1}} \left|m^{-1}\|Gx\|_2^2 - 1\right| > \varepsilon\right\} \le 2\exp(Cd - cm\varepsilon^2).$$
2. Deduce that, if $m \ge C\varepsilon^{-2}(k + \log(M/\rho))$, then with probability at least $1-\rho$,
$$(1-\varepsilon)\|x-y\|_2^2 \le m^{-1}\|G(x-y)\|_2^2 \le (1+\varepsilon)\|x-y\|_2^2 \quad \text{for every } x, y \in T.$$
Explain why controlling each V_j separately does not by itself prove this.
3. Apply the result to all vectors with at most s nonzero coordinates, $1 \le s \le n$, to obtain the sufficient bound
$$m \ge C\varepsilon^{-2}\left(s\log\frac{en}{s} + \log\frac{1}{\rho}\right).$$

Facts you may use. A unit sphere in dimension d has a $1/4$-net of size at most 9^d. For symmetric B and such a net $\mathcal{N}$, $\|B\|_{\mathrm{op}} \le 2\max_{x\in\mathcal{N}}|x^{\mathsf{T}}Bx|$. For part 2, consider the spaces $V_i + V_j$.

3. The ground-state energy of a Gaussian spin system

For $n \geq 2$, let $(Z_{ij})_{1 \leq j < i \leq n}$ be independent standard normal random variables. For $\sigma \in \{-1, 1\}^n$, define

$$H_\sigma = \sum_{i>j} Z_{ij} \sigma_i \sigma_j, \qquad E_n = \min_{\sigma \in \{-1,1\}^n} H_\sigma.$$

1. Prove that $\mathbb{E}E_n = -\mathbb{E} \max_\sigma H_\sigma$ and that $-\mathbb{E}E_n \leq Cn^{3/2}$.
2. Construct signs successively by taking $\sigma_1 = 1$ and choosing σ_i to maximize $\sigma_i \sum_{j<i} Z_{ij}\sigma_j$. Condition on the previous rows to show that the expected value attained is $\sqrt{2/\pi} \sum_{i=2}^n \sqrt{i-1}$. Deduce $cn^{3/2} \leq -\mathbb{E}E_n$.
3. Show that $Z \mapsto E_n$ is $\sqrt{n(n-1)/2}$-Lipschitz, and deduce

$$\mathbb{P}\left\{ \left| \frac{E_n}{\mathbb{E}E_n} - 1 \right| > \delta \right\} \leq 2e^{-c\delta^2 n} \qquad (\delta > 0).$$

   In particular, $E_n / \mathbb{E}E_n \to 1$ in probability.

Facts you may use. For centered Gaussian variables with variances at most v^2, $\mathbb{E} \max_{1 \leq j \leq N} X_j \leq v\sqrt{2 \log N}$, without any independence assumption. An L-Lipschitz function of a standard Gaussian vector satisfies $\mathbb{P}\{|f - \mathbb{E}f| > t\} \leq 2e^{-t^2/(2L^2)}$.

4. Noisy measurements and feasible recovery

Let $T \subset \mathbb{R}^n$ be nonempty and compact, let $x \in T$, and let A be an $m \times n$ standard Gaussian matrix. Observe $y = Ax + \eta$, where $\eta \in \mathbb{R}^m$ is deterministic. A reconstruction $\widehat{x} \in T$ may depend measurably on A and y.

You may use the following **Gaussian matrix deviation bound**: for every compact symmetric set $S \subset \mathbb{R}^n$ containing 0,

$$\mathbb{E} \sup_{h \in S} \left| \|Ah\|_2 - \sqrt{m}\|h\|_2 \right| \leq Cw(S).$$

No proof of this theorem is required.

1. Show that $w(T - T) = 2w(T)$. If $\|A\widehat{x} - y\|_2 \leq \delta$ almost surely, where $\delta \geq 0$ is deterministic, prove

$$\mathbb{E}\|\widehat{x} - x\|_2 \leq \frac{Cw(T) + \|\eta\|_2 + \delta}{\sqrt{m}}.$$

   The proof must allow $\widehat{x}$ to be chosen after observing A.
2. Explain when exact interpolation ($\delta = 0$) is feasible. For a constrained least-squares minimizer over T, prove the bound with $2\|\eta\|_2$ in place of $\|\eta\|_2 + \delta$.
3. Specialize to $T = RB_1^n$, $R > 0$, to obtain $\mathbb{E}\|\widehat{x} - x\|_2 \leq [CR\sqrt{\log(2n)} + 2\|\eta\|_2]/\sqrt{m}$ for constrained least squares. Explain, by a kernel example when $m < n$, why a feasibility restriction on $\widehat{x}$ is essential.

Hint. Apply the deviation bound uniformly on $T - T$ before substituting $h = \widehat{x} - x$. For RB_1^n, identify the supremum defining its width.

5. Random sections of the octahedron

Let $1 \le d < n$, let A be a $d \times n$ standard Gaussian matrix, and let $E = \ker A$. Prove the Garnaev–Gluskin bound

$$\mathbb{E}\,\mathrm{diam}(B_1^n \cap E) \le C\sqrt{\frac{\log(en/d)}{d}},$$

where diameter is measured in Euclidean distance. The following steps make the problem independent of later lectures.

1. Put $T_r = B_1^n \cap rB_2^n$. If $n^{-1/2} \le r \le 1$, prove

   $$w(T_r) \le C\sqrt{\log(enr^2)}.$$

   You may use $\mathbb{E}(\sum_{i=1}^{s}(g_i^*)^2)^{1/2} \le C\sqrt{s\log(en/s)}$, where $g_1^* \ge \cdots \ge g_n^*$ are the decreasing rearrangement of $|g_1|, \dots, |g_n|$.
2. Write $\mathrm{rad}(S) = \sup_{x\in S}\|x\|_2$. Prove that if $\mathrm{rad}(T_r \cap E) \le r/2$, then $\mathrm{rad}(B_1^n \cap E) \le r/2$.
3. You may use the following **Gaussian deviation theorem**: for compact symmetric S containing 0 and $u \ge 0$, with probability at least $1 - 2e^{-u^2}$,

   $$\sup_{h\in S}\Big|\|Ah\|_2 - \sqrt{d}\|h\|_2\Big| \le C_0\big[w(S) + u\,\mathrm{rad}(S)\big].$$

   Use it with $S = T_r$, $u = a\sqrt{d}$ for a sufficiently small absolute $a > 0$, and $r = B\sqrt{\log(en/d)/d}$ for a sufficiently large absolute B. Handle $r \ge 1$ separately. Deduce the asserted **expectation** bound, including the contribution of the exceptional event.

Hint for part 1. Split $\langle g, x\rangle$ over the s largest coordinates of $|g|$ and their complement, with $s = \lceil r^{-2}\rceil$. The two contributions are at most $r(\sum_{i\le s}(g_i^*)^2)^{1/2}$ and g_{s+1}^*, respectively; the second term is zero when $s = n$.

6. Random approximation and entropy of a convex hull

Let $v_1, \dots, v_N \in B_2^n$, where $N \ge 1$, and set $K = \mathrm{conv}\{0, \pm v_1, \dots, \pm v_N\}$.

1. For each $x \in K$ and integer $m \ge 1$, construct independent random vectors $Y_1, \dots, Y_m \in \{0, \pm v_1, \dots, \pm v_N\}$ with $\mathbb{E}Y_i = x$. Prove

   $$\mathbb{E}\left\|m^{-1}\sum_{i=1}^{m} Y_i - x\right\|_2^2 \le m^{-1}.$$

2. Deduce, for $0 < \varepsilon \le 1$,

   $$\log N(K, \|\cdot\|_2, \varepsilon) \le \lceil \varepsilon^{-2}\rceil \log(2N+1).$$

   Explain why the approximating points belong to K and why the net works simultaneously for all $x \in K$.
3. Independently prove $w(K) \le C\sqrt{\log(2N)}$. Why does inserting just the entropy estimate from part 2 into Dudley's integral fail to give a finite upper bound? Combine it with $N(K, \varepsilon) \le (1 + 2/\varepsilon)^n$ to explain why the actual entropy integral is nevertheless finite for every fixed n.

Hint. For part 3, a linear functional attains its maximum over a finite convex hull at one of its vertices. Distinguish a divergent upper bound from a divergent integral.

7. Generic chaining on a binary tree

For a finite metric space (T, d), define

$$\gamma_2(T,d) = \inf_{(T_j)} \sup_{t\in T} \sum_{j\ge 0} 2^{j/2} d(t, T_j), \qquad |T_0| = 1, \quad |T_j| \le 2^{2^j} \ (j \ge 1).$$

You may use **Talagrand's theorem**: for a centered Gaussian process with canonical metric $d(s,t) = (\mathbb{E}|X_s - X_t|^2)^{1/2}$,

$$c\gamma_2(T,d) \le \mathbb{E}\max_{t\in T} X_t \le C\gamma_2(T,d).$$

Let $T = \{0,1\}^m$, $m \ge 1$, be the leaves of a binary tree of depth m. Attach independent standard Gaussian variables g_e to its edges and put $X_t = \sum_{e\in[\varnothing,t]} g_e$.

1. If $\ell(s,t)$ is the length of the common prefix of two leaves, prove $d(s,t)^2 = 2(m - \ell(s,t))$.
2. Construct an admissible sequence by representing every prefix of length $\min\{m, 2^j\}$ at level $j \ge 1$ by one leaf, and taking a single leaf at level 0. Prove directly that $\gamma_2(T,d) \le Cm$.
3. Starting at the root, repeatedly follow the edge with the larger Gaussian label. Use independence of unexplored edges to show that the expected value at the resulting leaf is $m/\sqrt{\pi}$. Deduce $\gamma_2(T,d) \ge cm$ and $cm \le \mathbb{E}\max_t X_t \le Cm$.
4. Prove
$$\mathbb{P}\left\{\left|\max_t X_t - \mathbb{E}\max_t X_t\right| > u\right\} \le 2e^{-u^2/(2m)}.$$
Explain why, for $m \ge 2$, the leaves cannot be treated as independent Gaussian variables.

Hint. If a, b are independent standard normals, $\max(a,b) = (a+b+|a-b|)/2$. In part 2, bound the nonzero sum by $C\sqrt{m}\sum_{2^j<m} 2^{j/2}$, keeping the level-zero term.

8. Product geometry and sub-Gaussian processes

Let $a_1, \dots, a_q > 0$ and integers $d_1, \dots, d_q \ge 1$ be given. In the orthogonal product $\mathbb{R}^{d_1} \times \cdots \times \mathbb{R}^{d_q}$, let

$$T = \prod_{j=1}^{q} a_j B_2^{d_j}$$

with its Euclidean metric. The definition and theorem in Problem 7 extend to compact Euclidean sets and their continuous Gaussian processes.

1. Prove
$$w(T) = \sum_{j=1}^{q} a_j \mathbb{E}\|g^{(j)}\|_2, \qquad c\sum_{j=1}^{q} a_j\sqrt{d_j} \le \gamma_2(T) \le C\sum_{j=1}^{q} a_j\sqrt{d_j},$$
where $g^{(j)}$ is standard Gaussian in $\mathbb{R}^{d_j}$.
2. Let $(Y_t)_{t\in T}$ be a centered process with almost surely continuous paths, $Y_0 = 0$, and
$$\mathbb{E}e^{\lambda(Y_s - Y_t)} \le \exp\left(\frac{L^2\lambda^2\|s-t\|_2^2}{2}\right) \quad (s,t\in T,\ \lambda\in\mathbb{R}).$$
Using the generic chaining bound, prove $\mathbb{E}\sup_{t\in T}|Y_t| \le CL\sum_j a_j\sqrt{d_j}$.
3. The diameter is $2(\sum_j a_j^2)^{1/2}$. Compare this product with the Euclidean ball of that same radius $(\sum_j a_j^2)^{1/2}$ in dimension $D = \sum_j d_j$. Show that bounding width only by this radius and D can lose an unbounded factor, using $q = 2$, $d_1 = 1$, $d_2 = N$, $a_1 = 1$, $a_2 = N^{-1}$.

Facts you may use. For a standard Gaussian vector in $\mathbb{R}^d$, $c\sqrt{d} \le \mathbb{E}\|g\|_2 \le \sqrt{d}$. The generic chaining bound gives $\mathbb{E}\sup_t|Y_t - Y_0| \le CL\gamma_2(T)$ under the hypotheses in part 2.

Problems 3–5 develop past exam prompts supplied by the instructor. Problems 4 and 5 are related to the noisy-recovery and random-section exercises in Roman Vershynin, *High-Dimensional Probability*; the necessary matrix deviation statements are supplied in full here.